

Commutativity of Discrete Convolution

Definition

The discrete convolution of two sequences $x[n]$ and $y[n]$ is defined as:

$$(x * y)[n] = \sum_{k=-\infty}^{\infty} x[k] y[n - k]$$

Theorem

Convolution is commutative, that is

$$x * y = y * x$$

Proof

Starting from the definition:

$$(x * y)[n] = \sum_{k=-\infty}^{\infty} x[k] y[n - k]$$

Introduce a change of variable:

$$m = n - k$$

Then

$$k = n - m$$

Since the index still ranges over all integers, the sum can be rewritten as:

$$(x * y)[n] = \sum_{m=-\infty}^{\infty} x[n - m] y[m]$$

Using the commutativity of multiplication:

$$(x * y)[n] = \sum_{m=-\infty}^{\infty} y[m] x[n - m]$$

But this is exactly the definition of the convolution of y and x :

$$(y * x)[n] = \sum_{m=-\infty}^{\infty} y[m] x[n - m]$$

Therefore

$$(x * y)[n] = (y * x)[n]$$
$$x * y = y * x$$